Warning or Error Number	Short Description	Long Description
312	Low Water Flow - BP2	Low water flow - Booster Pump 2:Please seek technical assistance
313	Low Water Flow - BP3	Low water flow - Booster Pump 3:Please seek technical assistance
314	Low Water Flow - BP4	Low water flow - Booster Pump 4:Please seek technical assistance
315	Low Water Flow - BP5	Low water flow - Booster Pump 5:Please seek technical assistance
316	Low Water Flow - BP6	Low water flow - Booster Pump 6:Please seek technical assistance
317	Spare	Spare - Spare:Please seek technical assistance
318	Spare	Spare - Spare:Please seek technical assistance
319	Spare	Spare - Spare:Please seek technical assistance
320	Spare	Spare - Spare:Please seek technical assistance
321	Spare	Spare - Spare:Please seek technical assistance
322	Automatic source shut off	Source timed out due to period of inactivity:
323	Drum spreader - timeout	Drum spreader motor timed out after running continously for 1 minute:If this alarm persists
324	Rewind spreader - timeout	Rewind spreader motor timed out after running continously for 1 minute:If this alarm persists
325	Rewind layarm spreader - timeout	Rewind layarm spreader motor timed out after running continously for 1 minute:If this alarm persists
326	Varibow rotation limit	Varibow at end stop proxy switch:Please seek technical assistance
327	Spare	Spare - Spare:Please seek technical assistance
328	Spare	Spare - Spare:Please seek technical assistance
329	Spare	Spare - Spare:Please seek technical assistance
330	Spare	Spare - Spare:Please seek technical assistance
331	Circuit breaker tripped	Web threader circuit breaker tripped - E06_0 Panel:Please seek technical assist-ance
332	E-Stop tripped	Crane E-Stop activated: Reset crane E-Stop
333	Circuit breaker tripped	110/240V supply breaker tripped - E06_0 Panel:Please seek technical assistance
334	Spare	Spare - Spare:Please seek technical assistance
335	Circuit breaker tripped	Blower circuit breaker tripped - E06 Panel:Please seek technical assistance
336	Spare	Spare - Spare:Please seek technical assistance
337	Spare	Spare - Spare:Please seek technical assistance
338	Spare	Spare - Spare:Please seek technical assistance
339	Rewind chucks not closed	Rewind chucks are not fully closed or proxy switches not being made:
340	Unwind chucks not closed	Unwind chucks are not fully closed or proxy switches not being made:
341	Varibow checkback failed	Varibow spreader contactor checkback failed:Please seek technical assistance
342	Varibow PSU tripped	Varibow spreader 72V PSU tripped:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
343	Varibow circuit breaker tripped	Varibow spreader circuit breaker tripped:Please seek technical assistance
344	Varibow drive failed	Varibow spreader drive failed:Please seek technical assistance
345	Spare	Spare - Spare:Please seek technical assistance
346	Spare	Spare - Spare:Please seek technical assistance
347	Cryogenic fault - CG4	Cryogenic fault - CG4:Please seek technical assistance
348	Main water pressure low	Main feed or differential pressure low:Pumps will turn off after 5 minutes or immediately if feed pressure falls below 3 bar or differential falls below 2.5 bar
349	Main water pressure high	Main feed or differential pressure high:Please seek technical assistance
350	Mech water pressure low	Mech feed or differential pressure low:Please seek technical assistance
351	Mech water pressure high	Mech feed or differential pressure high:Please seek technical assistance
352	Spare	Spare - Spare:Please seek technical assistance
353	Profibus node failure	Profibus node failure - 1:Please seek technical assistance
354	Profibus node failure	Profibus node failure - 2:Please seek technical assistance
355	Profibus node failure	Profibus node failure - 3:Please seek technical assistance
356	Profibus node failure	Profibus node failure - 4:Please seek technical assistance
357	Profibus node failure	Profibus node failure - 5:Please seek technical assistance
358	Profibus node failure	Profibus node failure - 6 - Thyristor stack:Please seek technical assistance
359	Profibus node failure	Profibus node failure - 7 - Thyristor stack:Please seek technical assistance
360	Profibus node failure	Profibus node failure - 8 - Thyristor stack:Please seek technical assistance
361	Profibus node failure	Profibus node failure - 9 - Thyristor stack:Please seek technical assistance
362	Profibus node failure	Profibus node failure - 10 - Thyristor stack:Please seek technical assistance
363	Profibus node failure	Profibus node failure - 11 - Thyristor stack:Please seek technical assistance
364	Profibus node failure	Profibus node failure - 12 - Thyristor stack:Please seek technical assistance
365	Profibus node failure	Profibus node failure - 13 - Thyristor stack:Please seek technical assistance
366	Profibus node failure	Profibus node failure - 14 - Thyristor stack:Please seek technical assistance
367	Profibus node failure	Profibus node failure - 15 - Chiller:Please seek technical assistance
368	Profibus node failure	Profibus node failure - 16:Please seek technical assistance
369	Profibus node failure	Profibus node failure - 17 - Chiller:Please seek technical assistance
370	Profibus node failure	Profibus node failure - 18 - Wire feeders - Wirefeeders:Please seek technical assistance
371	Profibus node failure	Profibus node failure - 19 - Wire feeders - Wirefeeders:Please seek technical assistance
372	Profibus node failure	Profibus node failure - 20 - Pre Treater MFC1:Please seek technical assistance
373	Profibus node failure	Profibus node failure - 21 - Pre Treater MFC2:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
374	Profibus node failure	Profibus node failure - 22 - Pre Treater MFC3:Please seek technical assistance
375	Profibus node failure	Profibus node failure - 23 - Post Treater MFC1:Please seek technical assistance
376	Profibus node failure	Profibus node failure - 24 - Post Treater MFC2:Please seek technical assistance
377	Profibus node failure	Profibus node failure - 25 - Post Treater MFC3 - Post Treater MFC2:Please seek technical assistance
378	Profibus node failure	Profibus node failure - 26 - Drum Gas Wedge MFC1:Please seek technical assistance
379	Profibus node failure	Profibus node failure - 27 - Drum Gas Wedge MFC2:Please seek technical assistance
380	Profibus node failure	Profibus node failure - 28 - Rewind Gas Wedge MFC1:Please seek technical assistance
381	Profibus node failure	Profibus node failure - 29 - Rewind Gas Wedge MFC2:Please seek technical assistance
382	Profibus node failure	Profibus node failure - 30 - O2 MFC1:Please seek technical assistance
383	Profibus node failure	Profibus node failure - 31 - O2 MFC2:Please seek technical assistance
384	Profibus node failure	Profibus node failure - 32 - O2 MFC3:Please seek technical assistance
385	Profibus node failure	Profibus node failure - 33 - O2 MFC4:Please seek technical assistance
386	Profibus node failure	Profibus node failure - 34 - O2 MFC5:Please seek technical assistance
387	Profibus node failure	Profibus node failure - 35 - O2 MFC6:Please seek technical assistance
388	Profibus node failure	Profibus node failure - 36 - O2 MFC7:Please seek technical assistance
389	Profibus node failure	Profibus node failure - 37 - O2 MFC8:Please seek technical assistance
390	Profibus node failure	Profibus node failure - 38 - O2 MFC9:Please seek technical assistance
391	Profibus node failure	Profibus node failure - 39 - O2 MFC10:Please seek technical assistance
392	Profibus node failure	Profibus node failure - 40 - O2 MFC11:Please seek technical assistance
393	Profibus node failure	Profibus node failure - 41 - O2 MFC12:Please seek technical assistance
394	Profibus node failure	Profibus node failure - 42 - O2 MFC13:Please seek technical assistance
395	Profibus node failure	Profibus node failure - 43 - O2 MFC14:Please seek technical assistance
396	Profibus node failure	Profibus node failure - 44 - O2 MFC15:Please seek technical assistance
397	Profibus node failure	Profibus node failure - 45 - O2 MFC16:Please seek technical assistance
398	Profibus node failure	Profibus node failure - 46 - O2 MFC17:Please seek technical assistance
399	Profibus node failure	Profibus node failure - 47 - O2 MFC18:Please seek technical assistance
400	Profibus node failure	Profibus node failure - 48 - O2 MFC19:Please seek technical assistance
401	Profibus node failure	Profibus node failure - 49 - O2 MFC20:Please seek technical assistance
402	Profibus node failure	Profibus node failure - 50 - O2 MFC21:Please seek technical assistance
403	Profibus node failure	Profibus node failure - 51 - O2 MFC22:Please seek technical assistance
404	Profibus node failure	Profibus node failure - 52 - O2 MFC23:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
405	Profibus node failure	Profibus node failure - 53 - O2 MFC24:Please seek technical assistance
406	Profibus node failure	Profibus node failure - 54 - O2 MFC25:Please seek technical assistance
407	Profibus node failure	Profibus node failure - 55 - O2 MFC26:Please seek technical assistance
408	Profibus node failure	Profibus node failure - 56 - O2 MFC27:Please seek technical assistance
409	Profibus node failure	Profibus node failure - 57 - O2 MFC28:Please seek technical assistance
410	Profibus node failure	Profibus node failure - 58 - O2 MFC29:Please seek technical assistance
411	Profibus node failure	Profibus node failure - 59 - O2 MFC30:Please seek technical assistance
412	Profibus node failure	Profibus node failure - 60 - O2 MFC31:Please seek technical assistance
413	Profibus node failure	Profibus node failure - 61 - O2 MFC32:Please seek technical assistance
414	Profibus node failure	Profibus node failure - 62 - O2 MFC33:Please seek technical assistance
415	Profibus node failure	Profibus node failure - 63 - O2 MFC34:Please seek technical assistance
416	Profibus node failure	Profibus node failure - 64 - O2 MFC35:Please seek technical assistance
417	Profibus node failure	Profibus node failure - 65 - O2 MFC36:Please seek technical assistance
418	Profibus node failure	Profibus node failure - 66 - O2 MFC37:Please seek technical assistance
419	Profibus node failure	Profibus node failure - 67 - O2 MFC38:Please seek technical assistance
420	Profibus node failure	Profibus node failure - 68 - O2 MFC39:Please seek technical assistance
421	Profibus node failure	Profibus node failure - 69 - O2 MFC40:Please seek technical assistance
422	Profibus node failure	Profibus node failure - 70 - O2 MFC41:Please seek technical assistance
423	Profibus node failure	Profibus node failure - 71 - O2 MFC42:Please seek technical assistance
424	Profibus node failure	Profibus node failure - 72 - O2 MFC43:Please seek technical assistance
425	Profibus node failure	Profibus node failure - 73 - O2 MFC44:Please seek technical assistance
426	Profibus node failure	Profibus node failure - 74 - O2 MFC45:Please seek technical assistance
427	Profibus node failure	Profibus node failure - 75 - O2 MFC46:Please seek technical assistance
428	Profibus node failure	Profibus node failure - 76 - O2 MFC47:Please seek technical assistance
429	Profibus node failure	Profibus node failure - 77 - O2 MFC48:Please seek technical assistance
430	Profibus node failure	Profibus node failure - 78 - O2 MFC49:Please seek technical assistance
431	Profibus node failure	Profibus node failure - 79 - O2 MFC50:Please seek technical assistance
432	Profibus node failure	Profibus node failure - 80 - O2 MFC51:Please seek technical assistance
433	Profibus node failure	Profibus node failure - 81 - O2 MFC52:Please seek technical assistance
434	Profibus node failure	Profibus node failure - 82 - O2 MFC53:Please seek technical assistance
435	Profibus node failure	Profibus node failure - 83 - O2 MFC54:Please seek technical assistance
436	Profibus node failure	Profibus node failure - 84 - O2 MFC55:Please seek technical assistance
437	Profibus node failure	Profibus node failure - 85 - O2 MFC56:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
438	Profibus node failure	Profibus node failure - 86 - O2 MFC57:Please seek technical assistance
439	Profibus node failure	Profibus node failure - 87 - O2 MFC58:Please seek technical assistance
440	Profibus node failure	Profibus node failure - 88 - O2 MFC59:Please seek technical assistance
441	Profibus node failure	Profibus node failure - 89 - O2 MFC60:Please seek technical assistance
442	Profibus node failure	Profibus node failure - 90 - O2 MFC61:Please seek technical assistance
443	Profibus node failure	Profibus node failure - 91 - O2 MFC62:Please seek technical assistance
444	Profibus node failure	Profibus node failure - 92 - O2 MFC63:Please seek technical assistance
445	Profibus node failure	Profibus node failure - 93 - O2 MFC64:Please seek technical assistance
446	Profibus node failure	Profibus node failure - 94 - O2 MFC65:Please seek technical assistance
447	Profibus node failure	Profibus node failure - 95 - O2 MFC66:Please seek technical assistance
448	Profibus node failure	Profibus node failure - 96 - O2 MFC67:Please seek technical assistance
449	Profibus node failure	Profibus node failure - 97 - O2 MFC68:Please seek technical assistance
450	Profibus node failure	Profibus node failure - 98 - O2 MFC69:Please seek technical assistance
451	Profibus node failure	Profibus node failure - 99:Please seek technical assistance
452	Profibus node failure	Profibus node failure - 100:Please seek technical assistance
453	Profibus node failure	Profibus node failure - 101:Please seek technical assistance
454	Profibus node failure	Profibus node failure - 102:Please seek technical assistance
455	Profibus node failure	Profibus node failure - 103:Please seek technical assistance
456	Profibus node failure	Profibus node failure - 104:Please seek technical assistance
457	Profibus node failure	Profibus node failure - 105:Please seek technical assistance
458	Profibus node failure	Profibus node failure - 106:Please seek technical assistance
459	Profibus node failure	Profibus node failure - 107:Please seek technical assistance
460	Profibus node failure	Profibus node failure - 108:Please seek technical assistance
461	Profibus node failure	Profibus node failure - 109:Please seek technical assistance
462	Profibus node failure	Profibus node failure - 110:Please seek technical assistance
463	Profibus node failure	Profibus node failure - 111:Please seek technical assistance
464	Profibus node failure	Profibus node failure - 112:Please seek technical assistance
465	Profibus node failure	Profibus node failure - 113:Please seek technical assistance
466	Profibus node failure	Profibus node failure - 114:Please seek technical assistance
467	Profibus node failure	Profibus node failure - 115:Please seek technical assistance
468	Profibus node failure	Profibus node failure - 116:Please seek technical assistance
469	Profibus node failure	Profibus node failure - 117:Please seek technical assistance
470	Profibus node failure	Profibus node failure - 118:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
471	Profibus node failure	Profibus node failure - 119:Please seek technical assistance
472	Profibus node failure	Profibus node failure - 120:Please seek technical assistance
473	Profibus node failure	Profibus node failure - 121:Please seek technical assistance
474	Profibus node failure	Profibus node failure - 122:Please seek technical assistance
475	Profibus node failure	Profibus node failure - 123:Please seek technical assistance
476	Profibus node failure	Profibus node failure - 124:Please seek technical assistance
477	Profibus node failure	Profibus node failure - 125:Please seek technical assistance
478	Profibus node failure	Profibus node failure - 126:Please seek technical assistance
479	Profibus node failure	Profibus node failure - 127:Please seek technical assistance
480	Profibus node failure	Profibus node failure - 128:Please seek technical assistance
481	Spare	Spare - Spare:Please seek technical assistance
482	Water/Air Services not available	Water/Air services not available - Mech Traverse:Please seek technical assistance
483	Door open - cooling disabled	Door open - cooling disabled - E03_0 panel:Please seek technical assistance
484	Panel cooling fault	Panel cooling fault - E06_3 Panel:Please seek technical assistance
485	Door open - cooling disabled	Door open - cooling disabled - E06_3 Panel:Please seek technical assistance
486	Door open - cooling disabled	Door open - cooling disabled - E04 Panel:Please seek technical assistance
487	Door open - cooling disabled	Door open - cooling disabled - E06_0 Panel:Please seek technical assistance
488	Spare	Spare - Spare:Please seek technical assistance
489	Drive Fault	Fault - Draw roller:Please seek technical assistance
490	Profibus fault	Profibus fault - Draw roller:Please seek technical assistance
491	Drive fault	Drive fault - Draw roller:Please seek technical assistance
492	Blower fault	Blower fault - Draw roller:Please seek technical assistance
493	Roller slipping	Draw roller is slipping:
494	Communication failed	Communications failed - Drives PLC:Please seek technical assistance
495	Unwind sidelay proxy fault	Unwind sidelay proxy switch fault:Please seek technical assistance
496	Rewind sidelay proxy fault	Rewind sidelay proxy switch fault:Please seek technical assistance
497	Coolant Temperature - RP1	Coolant temperature warning - Rotary Pump 1:Please seek technical assistance
498	Coolant Temperature - RP1	Coolant temperature alarm - Rotary Pump 1:Please seek technical assistance
499	Cylinder Temperature - RP1	Cylinder temperature warning - Rotary Pump 1:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
500	Cylinder Temperature - RP1	Cylinder temperature alarm - Rotary Pump 1:Please seek technical assistance
501	Coolant Temperature - RP2	Coolant temperature warning - Rotary Pump 2:Please seek technical assistance
502	Coolant Temperature - RP2	Coolant temperature alarm - Rotary Pump 2:Please seek technical assistance
503	Cylinder Temperature - RP2	Cylinder temperature warning - Rotary Pump 2:Please seek technical assistance
504	Cylinder Temperature - RP2	Cylinder temperature alarm - Rotary Pump 2:Please seek technical assistance
505	Coolant Temperature - RP4	Coolant temperature warning - Rotary Pump 4:Please seek technical assistance
506	Coolant Temperature - RP4	Coolant temperature alarm - Rotary Pump 4:Please seek technical assistance
507	Cylinder Temperature - RP4	Cylinder temperature warning - Rotary Pump 4:Please seek technical assistance
508	Cylinder Temperature - RP4	Cylinder temperature alarm - Rotary Pump 4:Please seek technical assistance
509	Coolant Temperature - RP6	Coolant temperature warning - Rotary Pump 6:Please seek technical assistance
510	Coolant Temperature - RP6	Coolant temperature alarm - Rotary Pump 6:Please seek technical assistance
511	Cylinder Temperature - RP6	Cylinder temperature warning - Rotary Pump 6:Please seek technical assistance
512	Cylinder Temperature - RP6	Cylinder temperature alarm - Rotary Pump 6:Please seek technical assistance
513	Argon pressure failed to decay	Argon pressure failed to decay:Please seek technical assistance
514	Argon low supply pressure	Argon low supply pressure:Please seek technical assistance
515	Argon low pressure	Argon low pressure:Please seek technical assistance
516	Nitrogen pressure failed to decay	Nitrogen pressure failed to decay:Please seek technical assistance
517	Nitrogen low supply pressure	Nitrogen low supply pressure:Please seek technical assistance
518	Nitrogen low pressure	Nitrogen low pressure:Please seek technical assistance
519	Oxygen pressure failed to decay	Oxygen pressure failed to decay:Please seek technical assistance
520	Oxygen low supply pressure	Oxygen low supply pressure:Please seek technical assistance
521	O2V1 test failed	O2V1 test failed:Please seek technical assistance
522	O2V1 test failed	O2V1 test failed:Please seek technical assistance
523	Oxygen low pressure	Oxygen low pressure:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
524	Valve test pressure fail to decay	Valve test pressure fail to decay:Please seek technical assistance
525	Purge not completed	Purge not completed:Please seek technical assistance
526	Spare	Spare:Please seek technical assistance
527	Override on safety not act-ive	Override on safety not active:Please seek technical assistance
528	Acknowledge required	Acknowledge required:Please seek technical assistance
529	Emergency stop input fault	Emergency stop input fault:Please seek technical assistance
530	Mechanism traverse closed input fault	Mechanism traverse closed input fault:Please seek technical assistance
531	RP1 airflow switch fault - RP1	RP1 airflow switch fault:Please seek technical assistance
532	RP2 airflow switch fault - RP2	RP2 airflow switch fault:Please seek technical assistance
533	ARPS1 input fault	Argon Pressure Switch 1 ARPS1 input fault:Please seek technical assistance
534	ARPS2 input fault	Argon Pressure Switch 2 ARPS2 input fault:Please seek technical assistance
535	O2PS1 input fault	O2 Pressure Switch 1 O2PS1 input fault:Please seek technical assistance
536	O2PS2 input fault	O2 Pressure Switch 2 O2PS2 input fault:Please seek technical assistance
537	N2V1 output fault	N2 Valve 1 N2V1 output fault:Please seek technical assistance
538	N2PS1 input fault	N2 Pressure Switch 1 input fault:Please seek technical assistance
539	N2PS2 input fault	N2 Pressure Switch 2 N2PS2 input fault:Please seek technical assistance
540	O2V1 output fault	O2 Valve 1 O2V1 output fault:Please seek technical assistance
541	O2V2 output fault	O2 Valve 2 O2V2 output fault:Please seek technical assistance
542	ARV1 output fault	Argon Valve 1 ARV1 output fault:Please seek technical assistance
543	Watchdog fault	Watchdog fault:Please seek technical assistance
544	Spare	Spare:Please seek technical assistance
545	Softstart fault	Softstart fault - Rotary Pump 1:Please seek technical assistance
546	Softstart fault	Softstart fault - Rotary Pump 2:Please seek technical assistance
547	Softstart fault	Softstart fault - Rotary Pump 4:Please seek technical assistance
548	Softstart fault	Softstart fault - Rotary Pump 6:Please seek technical assistance
549	Gas 3 failed to decay	Gas 3 failed to decay - Precoat plasma treater:Please seek technical assistance
550	Gas 3 failed to decay	Gas 3 failed to decay - Postcoat plasma treater:Please seek technical assistance
551	Valve fault	Valve fault - Vacuum Valve VV11:Please seek technical assistance
552	Spare	Spare - Spare:Please seek technical assistance
553	Gas flow failure	Gas flow failure - Postcoat plasma treater:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
554	Gas 1 failed to decay	Gas 1 failed to decay - Precoat plasma treater:Please seek technical assistance
555	Gas 2 failed to decay	Gas 2 failed to decay - Precoat plasma treater:Please seek technical assistance
556	Gas 1 failed to decay	Gas 1 failed to decay - Postcoat plasma treater:Please seek technical assistance
557	Gas 2 failed to decay	Gas 2 failed to decay - Postcoat plasma treater:Please seek technical assistance
558	Spare	Spare - Spare:Please seek technical assistance
559	Spare	Spare - Spare:Please seek technical assistance
560	Spare	Spare - Spare:Please seek technical assistance
561	O2 MFC oxygen failed to decay	O2 MFC oxygen failed to decay - O2 system:Please seek technical assistance
562	O2 MFC oxygen overflow	O2 MFC oxygen overflow - O2 system:Please seek technical assistance
563	OD out of range warning	OD out of range warning - O2 system:Please seek technical assistance
564	OD out of range fault	OD out of range fault - O2 system:Please seek technical assistance
565	Water flow failed	Water flow failed - Postcoat plasma treater:Please seek technical assistance
566	Spare	Spare - Spare:Please seek technical assistance
567	Hawkeye comms fault	Communication fault on Hawkeye monitoring beam:Please seek technical assistance
568	Circuit breaker tripped	Circuit breaker tripped - Precoat plasma treater E07 Panel:Please seek technical assistance
569	Out of position	Out of position - Postcoat plasma treater:Please seek technical assistance
570	Drive fault	Drive fault - Postcoat plasma treater:Please seek technical assistance
571	Low water flow	Low water flow - Postcoat plasma treater:Please seek technical assistance
572	Low water flow	Low water flow - Precoat plasma treater:Please seek technical assistance
573	Regulation failed	Regulation failed - Precoat plasma treater:Please seek technical assistance
574	Regulation failed	Regulation failed - Postcoat plasma treater:Please seek technical assistance
575	Water flow failed	Water flow failed - Precoat plasma treater:Please seek technical assistance
576	Circuit breaker tripped	Circuit breaker tripped - Postcoat plasma treater E07 Panel:Please seek technical assistance
577	Failed to run	Failed to run - Postcoat plasma treater:Please seek technical assistance
578	Over temperature	Over temperature - Postcoat plasma treater:Please seek technical assistance
579	Oxygen concentration high	Oxygen concentration high - Postcoat plasma treater:Please seek technical assist-ance
580	Temperature fault	Temperature sensor 1 faulty on source shield
581	Temperature fault	Temperature sensor 2 faulty on source shield
582	Temperature fault	Temperature sensor 3 faulty on source shield
583	Temperature fault	Temperature sensor 4 faulty on source shield
584	Temperature fault	Source shield temperature high on source shield

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
585	Valve fault - VV10C	Valve fault - Vacuum Valve VV10C:Please seek technical assistance
586	Wire feed PSU A fault	Wire feed PSU A fault - E03_2 panel:Please seek technical assistance
587	Wire feed PSU B fault	Wire feed PSU B fault - E03_2 panel:Please seek technical assistance
588	Wire feed PSU C fault	Wire feed PSU C fault - E03_2 panel:Please seek technical assistance
589	Wire feed PSU D fault	Wire feed PSU D fault - E03_2 panel:Please seek technical assistance
590	Cyrogenic internal fault - CG1	Cyrogenic internal fault - CG1:Please seek technical assistance
591	Cyrogenic internal fault - CG2	Cyrogenic internal fault - CG2:Please seek technical assistance
592	Cyrogenic internal fault - CG3	Cyrogenic internal fault - CG3:Please seek technical assistance
593	Cyrogenic internal fault - CG4	Cyrogenic internal fault - CG4:Please seek technical assistance
594	Circuit breaker tripped	Circuit breaker tripped - CG4:Please seek technical assistance
595	Boat conduction check failed	Boat conduction check failed:
596	Circuit breaker tripped	220V circuit breaker tripped - E03_2 panel:Please seek technical assistance
597	Valve fault	Valve fault - Vacuum Valve VV12:Please seek technical assistance
598	OD out of O2 range	OD out of O2 range - O2 system:Please seek technical assistance
599	Auto tensioning failed	Auto tensioning failed:Please reset layarms
600	Over temperature	Over temperature - Precoat plasma treater:Please seek technical assistance
601	Rewind Home switch fault	Rewind home switch fault:Please seek technical assistance
602	Unwind Home switch fault	Unwind home switch fault:
603	Low water flow - BP1	Low water flow warning - Booster Pump 1:Please seek technical assistance
604	Low water flow - BP2	Low water flow warning - Booster Pump 2:Please seek technical assistance
605	Low water flow - BP3	Low water flow warning - Booster Pump 3:Please seek technical assistance
606	Low water flow - BP4	Low water flow warning - Booster Pump 4:Please seek technical assistance
607	Low water flow - BP5	Low water flow warning - Booster Pump 5:Please seek technical assistance
608	Low water flow - BP6	Low water flow warning - Booster Pump 6:Please seek technical assistance
609	PT1 Fault - High Gas Pressure Circuit 1	PT1 Fault - High Gas Pressure Circuit 1 - GRE Chiller:Please seek technical assistance
610	PT2 Fault - Low Gas Pressure Circuit 1	PT2 Fault - Low Gas Pressure Circuit 1 - GRE Chiller:Please seek technical assistance
611	PT3 Fault - High Gas Pressure Circuit 2	PT3 Fault - High Gas Pressure Circuit 2 - GRE Chiller:Please seek technical assistance
612	PT4 Fault - Low Gas Pressure Circuit 2	PT4 Fault - Low Gas Pressure Circuit 2 - GRE Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
613	PT5 Fault - Glycol pressure	PT5 Fault - Glycol pressure - GRE Chiller:Please seek technical assistance
614	TT1 Fault - Glycol Temperature	TT1 Fault - Glycol temperature - GRE Chiller:Please seek technical assistance
615	Tank high level switch fault	Tank high level switch fault - GRE Chiller:Please seek technical assistance
616	Chiller fault	Chiller fault - GRE Chiller:Please seek technical assistance
617	Tank low level switch fault	Tank low level switch fault - GRE Chiller:Please seek technical assistance
618	Pump over current alarm	Pump over current alarm - GRE Chiller:Please seek technical assistance
619	Heater 1 over current	Heater 1 over current alarm - GRE Chiller:Please seek technical assistance
620	Heater 2 over current	Heater 2 over current alarm - GRE Chiller:Please seek technical assistance
621	Heater 3 over current	Heater 3 over current alarm - GRE Chiller:Please seek technical assistance
622	Compressor 1 over current	Compressor 1 over current alarm - GRE Chiller:Please seek technical assistance
623	Compressor 2 over current	Compressor 2 over current alarm - GRE Chiller:Please seek technical assistance
624	Compressor 3 over current	Compressor 3 over current alarm - GRE Chiller:Please seek technical assistance
625	Compressor 4 over current	Compressor 4 over current alarm - GRE Chiller:Please seek technical assistance
626	Compressor 1 protection module	Compressor 1 protection module alarm - GRE Chiller:Please seek technical assistance
627	Compressor 2 protection module	Compressor 2 protection module alarm - GRE Chiller:Please seek technical assistance
628	Compressor 3 protection module	Compressor 3 protection module alarm - GRE Chiller:Please seek technical assistance
629	Compressor 4 protection module	Compressor 4 protection module alarm - GRE Chiller:Please seek technical assistance
630	High pressure software trip circuit 1	High pressure software trip circuit 1 - GRE Chiller:Please seek technical assistance
631	Low pressure software trip circuit 1	Low pressure software trip circuit 1 - GRE Chiller:Please seek technical assistance
632	High pressure software trip circuit 2	High pressure software trip circuit 2 - GRE Chiller:Please seek technical assistance
633	Low pressure software trip circuit 2	Low pressure software trip circuit 2 - GRE Chiller:Please seek technical assistance
634	Primary circuit flow trip	Primary circuit flow trip - GRE Chiller:Please seek technical assistance
635	Glycol circuit flow trip	Glycol circuit flow trip - GRE Chiller:Please seek technical assistance
636	HPLP hard trip circuit 1	HPLP hard trip circuit 1 - GRE Chiller:Please seek technical assistance
637	HPLP hard trip circuit 2	HPLP hard trip circuit 2 - GRE Chiller:Please seek technical assistance
638	Supply fault - Phase fail/rotation	Supply fault - Phase fail / rotation alarm - GRE Chiller:Please seek technical assistance
639	Expansion valve driver alarm circuit 1	Expansion valve driver alarm circuit 1 - GRE Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
640	Expansion valve driver alarm circuit 2	Expansion valve driver alarm circuit 2 - GRE Chiller:Please seek technical assistance
641	Terminal fault stop circuit 1	Terminal fault stop circuit 1 - GRE Chiller:Please seek technical assistance
642	Terminal fault stop circuit 2	Terminal fault stop circuit 2 - GRE Chiller:Please seek technical assistance
643	Terminal fault	Terminal fault stop whole machine - GRE Chiller:Please seek technical assistance
644	Watchdog fault	Watchdog fault - GRE Chiller:Please seek technical assistance
645	TT4 - Temperature probe fault	TT4 - Temperature probe - GRE Chiller:Please seek technical assistance
646	TT5 - Temperature probe fault	TT5 - Temperature probe - GRE Chiller:Please seek technical assistance
647	TT6 - Temperature probe fault	TT6 - Temperature probe - GRE Chiller:Please seek technical assistance
648	Spare	Spare - GRE Chiller:Please seek technical assistance
649	Spare	Spare - GRE Chiller:Please seek technical assistance
650	Spare	Spare - GRE Chiller:Please seek technical assistance
651	Spare	Spare - GRE Chiller:Please seek technical assistance
652	Spare	Spare - GRE Chiller:Please seek technical assistance
653	Spare	Spare - GRE Chiller:Please seek technical assistance
654	Spare	Spare - GRE Chiller:Please seek technical assistance
655	Spare	Spare - GRE Chiller:Please seek technical assistance
656	Spare	Spare - GRE Chiller:Please seek technical assistance
657	Sensor break ch. 4 external	Sensor break ch. 4 external - GWK Chiller:Please seek technical assistance
658	Sensor break ch. 5 external	Sensor break ch. 5 external - GWK Chiller:Please seek technical assistance
659	Sensor break ch. 6 external	Sensor break ch. 6 external - GWK Chiller:Please seek technical assistance
660	Sensor break ch. 7 external	Sensor break ch. 7 external - GWK Chiller:Please seek technical assistance
661	Sensor rupture reserve	Sensor rupture reserve - GWK Chiller:Please seek technical assistance
662	Sensor rupture reserve	Sensor rupture reserve - GWK Chiller:Please seek technical assistance
663	Sensor rupture reserve	Sensor rupture reserve - GWK Chiller:Please seek technical assistance
664	Sensor rupture reserve	Sensor rupture reserve - GWK Chiller:Please seek technical assistance
665	Sensor break ch. 0 flow line	Sensor break ch. 0 flow line temperature operating circuit - 1B1 - GWK Chiller:Please seek technical assistance
666	Sensor break ch. 1 flow line	Sensor break ch. 1 flow line temperature operating circuit - 1B2 - GWK Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
667	Sensor break ch. 2 outlet evaporator	Sensor break ch. 2 outlet evaporator temperature - 1B3 - GWK Chiller:Please seek technical assistance
668	Sensor break ch. 3 reserve	Sensor break ch. 3 reserve - GWK Chiller:Please seek technical assistance
669	Sensor break ch. 0 external	Sensor break ch. 0 external - GWK Chiller:Please seek technical assistance
670	Sensor break ch. 1 external	Sensor break ch. 1 external - GWK Chiller:Please seek technical assistance
671	Sensor break ch. 2 external	Sensor break ch. 2 external - GWK Chiller:Please seek technical assistance
672	Sensor break ch. 3 external	Sensor break ch. 3 external - GWK Chiller:Please seek technical assistance
673	Fault expansion valve circuit 1	Fault expansion valve circuit 1 - GWK Chiller:Please seek technical assistance
674	Power failure heaters	Power failure heaters - GWK Chiller:Please seek technical assistance
675	Power failure drives	Power failure drives - GWK Chiller:Please seek technical assistance
676	Emergency Stop monitoring active	Emergency Stop monitoring active - GWK Chiller:Please seek technical assistance
677	Fault expansion valve circuit 2	Fault expansion valve circuit 2 - GWK Chiller:Please seek technical assistance
678	Spare	Spare - GWK Chiller:Please seek technical assistance
679	Release compressor fault INT 69 Circuit 1	Release compressor fault INT 69 Circuit 1 - GWK Chiller:Please seek technical assistance
680	Release compressor fault INT 69 Circuit 2	Release compressor fault INT 69 Circuit 2 - GWK Chiller:Please seek technical assistance
681	Fault frost protector monitor	Fault frost protector monitor - GWK Chiller:Please seek technical assistance
682	Fault flow monitor	Fault flow monitor - GWK Chiller:Please seek technical assistance
683	Spare	Spare - GWK Chiller:Please seek technical assistance
684	Dry-run protection container	Dry-run protection container - GWK Chiller:Please seek technical assistance
685	Check fill level	Check fill level - GWK Chiller:Please seek technical assistance
686	Spare	Spare - GWK Chiller:Please seek technical assistance
687	Overcurrent evaporator pump 1	Overcurrent evaporator pump 1 - GWK Chiller:Please seek technical assistance
688	Spare	Spare - GWK Chiller:Please seek technical assistance
689	Overcurrent compressor circuit 1	Overcurrent compressor circuit 1 - GWK Chiller:Please seek technical assistance
690	High pressure fault circuit 1	High pressure fault circuit 1 - GWK Chiller:Please seek technical assistance
691	Oil level fault circuit 1	Oil level fault circuit 1 - GWK Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
692	CoreSense Diagnostics cooling circuit 1	Collective error message. CoreSense Diagnostics cooling circuit 1 - GWK Chiller-:Please seek technical assistance
693	Low pressure fault	Low pressure fault - GWK Chiller:Please seek technical assistance
694	Oil flow fault circuit 1	Oil flow fault circuit 1 - GWK Chiller:Please seek technical assistance
695	High pressure fault (high pressure limiter)	High pressure fault (high pressure limiter) - GWK Chiller:Please seek technical assistance
696	Spare	Spare - GWK Chiller:Please seek technical assistance
697	Overcurrent compressor circuit 2	Overcurrent compressor circuit 2 - GWK Chiller:Please seek technical assistance
698	High pressure fault circuit 2	High pressure fault circuit 2 - GWK Chiller:Please seek technical assistance
699	Oil level fault circuit 2	Oil level fault circuit 2 - GWK Chiller:Please seek technical assistance
700	CoreSense Diagnostics cooling circuit 2	Collective error message. CoreSense Diagnostics cooling circuit 2 - GWK Chiller-:Please seek technical assistance
701	Spare	Spare - GWK Chiller:Please seek technical assistance
702	Oil flow fault circuit 2	Oil flow fault circuit 2 - GWK Chiller:Please seek technical assistance
703	Spare	Spare - GWK Chiller:Please seek technical assistance
704	Spare	Spare - GWK Chiller:Please seek technical assistance
705	Fault cooling circuit 1	Fault cooling circuit 1 - GWK Chiller:Please seek technical assistance
706	Fault cooling circuit 2	Fault cooling circuit 2 - GWK Chiller:Please seek technical assistance
707	Fault heating circuit pump	Fault heating circuit pump - GWK Chiller:Please seek technical assistance
708	Fault refeeding pump	Fault refeeding pump - GWK Chiller:Please seek technical assistance
709	Over Temperature	Overtemperature heater - GWK Chiller:Please seek technical assistance
710	Heater over current	Overcurrent heater - GWK Chiller:Please seek technical assistance
711	Spare	Spare - GWK Chiller:Please seek technical assistance
712	Spare	Spare - GWK Chiller:Please seek technical assistance
713	Data transmission ProfiNet disrupted	Data transmission ProfiNet disrupted - GWK Chiller:Please seek technical assistance
714	Please switch on one evaporator pump	Please switch on one evaporator pump - GWK Chiller:Please seek technical assistance
715	Time monitoring heat-up time compressor active	Time monitoring heat-up time compressor active - GWK Chiller:Please seek technical assistance
716	Spare	Spare - GWK Chiller:Please seek technical assistance
717	Spare	Spare - GWK Chiller:Please seek technical assistance
718	Spare	Spare - GWK Chiller:Please seek technical assistance
719	Spare	Spare - GWK Chiller:Please seek technical assistance
720	Spare	Spare - GWK Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
721	Chiller Max Level	IF Chiller Max Level - IF Chiller:Please seek technical assistance
722	Chiller General Overload	IF Chiller General Overload - IF Chiller:Please seek technical assistance
723	Chiller Evaporator flow switch fault	IF Chiller Evaporator flow switch fault - IF Chiller:Please seek technical assistance
724	Chiller Condenser flow switch fault	IF Chiller Condenser flow switch fault - IF Chiller:Please seek technical assistance
725	Chiller Evaporator pump overload	IF Chiller Evaporator pump overload - IF Chiller:Please seek technical assistance
726	Chiller pumps run hours exceed limit	IF Chiller pumps run hours exceed limit - IF Chiller:Please seek technical assistance
727	Chiller Ice alarm	IF Chiller Ice alarm - IF Chiller:Please seek technical assistance
728	Chiller Evaporator 2 ice alarm	IF Chiller Evaporator 2 ice alarm - IF Chiller:Please seek technical assistance
729	Chiller general high pres- sure	IF Chiller general high pressure alarm - IF Chiller:Please seek technical assistance
730	Chiller circuit 1 high pres- sure	IF Chiller circuit1 high pressure alarm - IF Chiller:Please seek technical assistance
731	Chiller circuit 2 high pres- sure	IF Chiller circuit2 high pressure alarm - IF Chiller:Please seek technical assistance
732	Chiller circuit 3 high pres- sure	IF Chiller circuit3 high pressure alarm - IF Chiller:Please seek technical assistance
733	Chiller circuit 4 high pres- sure	IF Chiller circuit4 high pressure alarm - IF Chiller:Please seek technical assistance
734	Chiller general low pres- sure	IF Chiller general low pressure alarm - IF Chiller:Please seek technical assistance
735	Chiller circuit 1 low pres- sure	IF Chiller circuit 1 low pressure alarm - IF Chiller:Please seek technical assistance
736	Chiller circuit 2 low pres- sure	IF Chiller circuit 2 low pressure alarm - IF Chiller:Please seek technical assistance
737	Chiller circuit 3 low pres- sure	IF Chiller circuit 3 low pressure alarm - IF Chiller:Please seek technical assistance
738	Chiller circuit 4 low pres- sure	IF Chiller circuit 4 low pressure alarm - IF Chiller:Please seek technical assistance
739	Chiller general com- pressor overload	IF Chiller general compressor overload - IF Chiller:Please seek technical assistance
740	Chiller circuit 1 com- pressor overload	IF Chiller circuit 1 compressor overload - IF Chiller:Please seek technical assistance
741	Chiller circuit 2 com- pressor overload	IF Chiller circuit 2 compressor overload - IF Chiller:Please seek technical assistance
742	Chiller circuit 3 com- pressor overload	IF Chiller circuit 3 compressor overload - IF Chiller:Please seek technical assistance
743	Chiller circuit 4 com- pressor overload	IF Chiller circuit 4 compressor overload - IF Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
744	Chiller circuit 5 compressor overload	IF Chiller circuit 5 compressor overload - IF Chiller:Please seek technical assistance
745	Chiller circuit 6 compressor overload	IF Chiller circuit 6 compressor overload - IF Chiller:Please seek technical assistance
746	Chiller fan anomaly	IF Chiller fan anomaly - IF Chiller:Please seek technical assistance
747	Chiller compressor 1 run hours exceed limit	IF Chiller compressor 1 run hours exceed limit - IF Chiller:Please seek technical assistance
748	Chiller compressor 2 run hours exceed limit	IF Chiller compressor 2 run hours exceed limit - IF Chiller:Please seek technical assistance
749	Chiller compressor 3 run hours exceed limit	IF Chiller compressor 3 run hours exceed limit - IF Chiller:Please seek technical assistance
750	Chiller compressor 4 run hours exceed limit	IF Chiller compressor 4 run hours exceed limit - IF Chiller:Please seek technical assistance
751	Chiller compressor 5 run hours exceed limit	IF Chiller compressor 5 run hours exceed limit - IF Chiller:Please seek technical assistance
752	Chiller compressor 6 run hours exceed limit	IF Chiller compressor 6 run hours exceed limit - IF Chiller:Please seek technical assistance
753	Chiller no filling water	IF Chiller no filling water - IF Chiller:Please seek technical assistance
754	Chiller heaters	
755	Chiller circulation pump Z1 overload	IF Chiller circulation pump Z1 overload - IF Chiller:Please seek technical assistance
756	Chiller circulation pump Z2 overload	IF Chiller circulation pump Z2 overload - IF Chiller:Please seek technical assistance
757	Chiller security alarm	IF Chiller security alarm - IF Chiller:Please seek technical assistance
758	Chiller Max loading time exceeded	IF Chiller Max loading time exceeded - IF Chiller:Please seek technical assistance
759	Chiller Phase Sequence error	IF Chiller Phase Sequence error - IF Chiller:Please seek technical assistance
760	Chiller Supply Probe fault	IF Chiller Supply Probe fault - IF Chiller:Please seek technical assistance
761	Chiller Condenser probe fault	IF Chiller Condenser probe fault - IF Chiller:Please seek technical assistance
762	Chiller antifrost probe fault	IF Chiller antifrost probe fault - IF Chiller:Please seek technical assistance
763	Chiller freecooling probe fault	IF Chiller freecooling probe fault - IF Chiller:Please seek technical assistance
764	Chiller supply pressure probe fault Z1	IF Chiller supply pressure probe fault Z1 - IF Chiller:Please seek technical assistance
765	Chiller thermo supply probe fault Z1	IF Chiller thermo supply probe fault Z1 - IF Chiller:Please seek technical assistance
766	Chiller remote probe fault Z1	IF Chiller remote probe fault Z1 - IF Chiller:Please seek technical assistance

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
767	Chiller remote setpoint probe fault Z1	IF Chiller remote setpoint probe fault Z1 - IF Chiller:Please seek technical assistance
768	Chiller security probe fault Z1	IF Chiller security probe fault Z1 - IF Chiller:Please seek technical assistance
769	Chiller supply pressure probe fault Z2	IF Chiller supply pressure probe fault Z2 - IF Chiller:Please seek technical assistance
770	Chiller thermo supply probe fault Z2	IF Chiller thermo supply probe fault Z2 - IF Chiller:Please seek technical assistance
771	Chiller remote probe fault Z2	IF Chiller remote probe fault Z2 - IF Chiller:Please seek technical assistance
772	Chiller remote setpoint probe fault Z2	IF Chiller remote setpoint probe fault Z2 - IF Chiller:Please seek technical assistance
773	Chiller security probe fault Z2	IF Chiller security probe fault Z2 - IF Chiller:Please seek technical assistance
774	Chiller High Temperature	IF Chiller High temperature alarm Z1 - IF Chiller:Please seek technical assistance
775	Chiller High Temperature	IF Chiller High temperature warning Z1 - IF Chiller:Please seek technical assistance
776	Chiller High Temperature	IF Chiller High temperature alarm Z2 - IF Chiller:Please seek technical assistance
777	Chiller High Temperature	IF Chiller High temperature warning Z2 - IF Chiller:Please seek technical assistance
778	Chiller slave connection alarm	IF Chiller slave connection alarm - IF Chiller:Please seek technical assistance
779	Chiller no master presence	IF Chiller no master presence - IF Chiller:Please seek technical assistance
780	Chiller watchdog profibus fault	IF Chiller watchdog profibus fault - IF Chiller:Please seek technical assistance
781	Chiller general alarm	IF Chiller general alarm - IF Chiller:Please seek technical assistance
782	Spare	Spare - IF Chiller:Please seek technical assistance
783	Spare	Spare - IF Chiller:Please seek technical assistance
784	Spare	Spare - IF Chiller:Please seek technical assistance
1892	Winder motor heat exchanger temperature fault	Heat exchanger temperature is too high
1893	Winder motor heat exchanger temperature warning	Heat exchanger temperature is too high
1894	Water cooled motor fault	A winder motor has over heated
1895	Winder motor heat exchange breaker tripped	Circuit breaker tripped
1896	Unwind sidelay movement took too long	A sidelay function took longer than expected
1897	Unwind chuck movement took too long	A chuck function took longer than expected

Table 9.2-1 List of Alarms

Warning or Error Number	Short Description	Long Description
1898	Low unwind chuck pressure	The hydraulic pressure at the unwinder is too low
1899	Edge guider fault	
1900	Rewind sidelay movement took too long	A sidelay function took longer than expected
1901	Rewind chuck movement took too long	A chuck function took longer than expected
1902	Low rewind chuck pressure	The hydraulic pressure at the unwinder is too low

Table 9.2-1 List of Alarms

Maintenance Warnings

In the situation where a maintenance task timer exceeds the time set by the maintenance technician.

A warning message will be shown on this screen.

This will have a PINK background, until it is acknowledged.

The warning message will stay present, until the timer is reset by the maintenance technician when he logs onto the system. See 9.2 HMI Generated Alarms and Warnings on page 384 for information.

9.3 HMI Status Messages

Note !
Not all these messages apply to all variants of this type of machine. It will depend upon optional equipment being installed. **Some messages are duplicates as they are used at different times in the operation sequence.**

Layarm Status Rewind

Message		Description
0	Idle with motor power off	With drive de-energised, for example, with G-stop off, this message will be shown.
1	Energise servo drive	Initial message, once the drive system has been reset.
2	Find home position	Indicates that drive is being instructed to move to the home switch position
3	Idle - at home, reel diameter not measured	Layarm has moved to the home position following a “Reset layarms” instruction
4	Move into reel for diameter measurement	Layarm is moving to the roll/core to the measure the roll diameter.
5	Diameter measurement complete	Diameter measurement step completed
6	Diameter measurement complete	Diameter measurement step completed
8	Finding home position	Indicates that drive is being instructed to move to the home switch position
9	Idle at home position and diameter measured	Layarm has moved to the roll/core, measured the roll diameter, and is now moved to the home position.
10	Position to lay-off	Indicated when layarm is being commanded to move to the lay-off (gap) position
11	At lay-off position	Indicated when layarm is at the lay-off (gap) position
12	Position to lay-on	Indicated when layarm is being commanded to move to the lay-on (contact) position on the roll.
13	At lay-on position	Indicated when layarm is at the lay-on (contact) position on the roll.
14	FAST move into reels(web-break or E-stop with vessel closed)	Indicated when web break or e-stop command is initialised.
15	Inch down	When using the rear pushbuttons, using the “OUT” button, to drive the arm away from the roll/core
16	Inch up	When using the rear pushbuttons, using the “IN” button, to drive the arm towards the roll/core

Table 9.3-1 Rewind Layarm Drive Status Messages

Message		Description
17	Idle and diameter measured	Layarm has moved to the roll/core, measured the roll diameter, and is now powered down with the holding brake engaged.
18	Inch down	When using the rear pushbuttons, using the “OUT” button, to drive the arm away from the roll/core
19	Inch up	When using the rear pushbuttons, using the “IN” button, to drive the arm towards the roll/core
20	Error moving to home position	Indicated when unable to reach the home position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen
21	Error when measuring diameters	Indicated when unable to reach the roll diameter position.
22	Return to home error from diameter measurement	Indicated when unable to reach the home position. Possiblereasons: • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen
23	Drive error during lay-on /lay-off positioning	Indicated when unable to reach the required position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Check drive diagnostic screen for any error codes, indicating why the drive has stopped.
24	Request manual diameter	Possible reasons: • Core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached • Travel limit has been set to a larger diameter which prevents the lay arm reaching normal sized rolls, for instance when side cheeks are fitted for slippery substrates.
25	Request manual diameter	Possible reasons: • Core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached • Travel limit has been set to a larger diameter which prevents the lay arm reaching normal sized rolls, for instance when side cheeks are fitted for slippery substrates.

Table 9.3-1 Rewind Layarm Drive Status Messages

Message		Description
26	Request manual diameter	Possible reasons: • Core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached • Travel limit has been set to a larger diameter which prevents the lay arm reaching normal sized rolls, for instance when side cheeks are fitted for slippery substrates.
30	Reset technology board and drive	Indicated on certain types of drive systems when first energised.
40	Change to positioning mode	Indicated briefly when changing between lay on (contact) and lay off (gap) modes.
41	Move to found home position	Indicated when system has been commanded to move to the “home “position on some drive systems.
42	Moving to found home position	Indicated when system is being commanded to move to the “home “position on some drive systems.

Table 9.3-1 Rewind Layarm Drive Status Messages

Layarm Status Unwind

Message		Description
0	Idle with motor power off	With drive de-energised, for example, with G-stop off, this message will be shown.
1	Energise servo drive	Initial message, once the drive system has been reset.
2	Find home position	Indicates that drive is being instructed to move to the home switch position
3	Idle - at home, reel diameter not measured	Layarm has moved to the home position following a “Reset layarms” instruction
4	Move into reel for diameter measurement	Layarm is moving to the roll/core to the measure the roll diameter.
5	Diameter measurement complete	Diameter measurement step completed
6	Diameter measurement complete	Diameter measurement step completed
8	Finding home position	Indicates that drive is being instructed to move to the home switch position
9	Idle at home position and diameter measured	Layarm has moved to the roll/core, measured the roll diameter, and is now moved to the home position.
10	Position to lay-off	Indicated when layarm is being commanded to move to the lay-off (gap) position
11	At lay-off position	Indicated when layarm is at the lay-off (gap) position

Table 9.3-2 Unwind Layarm Drive Status Messages

Message		Description
12	Position to lay-on	Indicated when layarm is being commanded to move to the lay-on (contact) position on the roll.
13	At lay-on position	Indicated when layarm is at the lay-on (contact) position on the roll.
14	FAST move into reels(web-break or E-stop with vessel closed)	Indicated when web break or e-stop command is initialised.
15	Inch down	When using the rear pushbuttons, using the “OUT” button, to drive the arm away from the roll/core
16	Inch up	When using the rear pushbuttons, using the “IN” button, to drive the arm towards the roll/core
17	Idle and diameter measured	Layarm has moved to the roll/core, measured the roll diameter, and is now powered down with the holding brake engaged.
18	Inch down	When using the rear pushbuttons, using the “OUT” button, to drive the arm away from the roll/core
19	Inch up	When using the rear pushbuttons, using the “IN” button, to drive the arm towards the roll/core
20	Error moving to home position	Indicated when unable to reach the home position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen
21	Error when measuring diameters	Indicated when unable to reach the roll diameter position. Possible Reasons:-
22	Return to home error from diameter measurement	Indicated when unable to reach the home position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen
23	Drive error during lay-on /lay-off positioning	Indicated when unable to reach the required position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Check drive diagnostic screen for any error codes, indicating why the drive has stopped.

Table 9.3-2 Unwind Layarm Drive Status Messages

Message		Description
24	Request manual diameter	Indicated when core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached.
25	Request manual diameter	Indicated when core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached.
26	Request manual diameter	Indicated when core diameter is less than 180mm and the programmed limit of movement from the home switch has been reached.
30	Reset technology board and drive	Indicated on certain types of drive systems when first energised.
40	Change to positioning mode	Indicated briefly when changing between lay on (contact) and lay off (gap) modes.
41	Move to found home position	Indicated when system has been commanded to move to the “home “position on some drive systems.
42	Moving to found home position	Indicated when system is being commanded to move to the “home “position on some drive systems.

Table 9.3-2 Unwind Layarm Drive Status Messages

Figure 9.3-1 Unwind Layarm Drive Status Messages

Plasma Treater Process Gas Supply

Message		Description
0	Flow	Control mode selected for fixed flow rate at the sccm value entered. The limit is the rating of the MFC.
1	Pressure	Control mode selected to flow gas to the pressure control set-point within the treater as measured by the Baratron gauge.

Table 9.3-3 Plasma Treater Process Gas supply status messages

Inline Deposition Monitor Mode - Diagnostic Screen Command

Message		Description
0	No Action	No Action
1	Calibrate	Start calibrate process
2	Self-Test	Self-test the deposition system, when switching evaporator power on.
3	Force Self-Test	Self-test mode forced by operator
4	Initialise	Downloading deposition set points to control system.
5	Acknowledge	Acknowledge any command
6	Reset LROD Gain	Rest all gain settings to default value on deposition system.
7	RF Up	Increase RF tuning signal on non-contact deposition system.
8	RF Down	Decrease RF tuning signal on non-contact deposition system.

Table 9.3-4 Inline deposition monitor status messages

Drive system Autoloss sequence

Many of these status messages can apply to all rollers that this function is carried out on.

Message		Description
0	Idle	Function not active.
1	Warm-up in progress	Indicated when the warm up status is in progress. A timer counts down to indicate how long is left.
2 - 12	Compensation speed 1- 12 active - Forward	Indicates which stage in the sequence the operation is at.
13	Auto loss tuning complete	Indicates that the autos sequence has been completed
14	Ramp to zero speed - Reverse speed	Indicates which stage in the sequence the operation is at.
15-25	Compensation speed 1 – 11 active - Reverse	Indicates which stage in the sequence the operation is at.
26	Auto loss tuning complete	Indicates that the autos sequence has been completed
27	Ramp to zero speed - sequence aborted	Indicated, if user aborts the autoloss sequence.
49	Internal/External E-stop active	External E-stop activated while sequence had not been completed.
50	Unwind/rewind diameter exceeds safe	Roll diameters exceed maximum diameter parameter. This is normally 1mm larger than the machine specification.
51	Auto loss compensation not enabled	Indicated if one of the following interlocks have not been met. • High vacuum (VV7) valves open • Chamber closed • Layarms have completed get diameter sequence and are home position.
52	Vessel not closed - unable to start	Indicates that the chamber is not closed, as this is one of the interlocks required to allow the sequence to take place.
53	Drive fault occurred during warm-up	Message will be indicated if a fault occurs. Reference should be made to the drive diagnostic screen at this time for an error code to allow further troubleshooting.
54-64	Drive fault occurred during speed 1-11	Message will be indicated if a fault occurs. Reference should be made to the drive diagnostic screen at this time for an error code to allow further troubleshooting.
65	Drive not running	Indicated when drive is not running.
70	Drive failed to reach zero speed	Indicated, if user aborts the autoloss sequence.

Table 9.3-5 Drive system autolosses sequence status messages

Message		Description
71-81	Drive fault occurred during speed 1-11	Message will be indicated if a fault occurs. Reference should be made to the drive diagnostic screen at this time for an error code to allow further troubleshooting.

Table 9.3-5 Drive system autolosses sequence status messages

Plasma system ramp function messages

Message		Description
0	Fixed	The rate of ramp of the power set-point will be fixed
1	Ramped	The rate of ramp of the power set-point will be ramped against line speed

Table 9.3-6 Plasma treater control status messages

Selection of plasma system process gases messages

Message		Description
0		No gas selected
1	Oxygen	Oxygen gas selected
2	Nitrogen	Nitrogen gas selected
3	Air	Air gas selected
4	Pre-mix	Pre-mix gas selected
5	Argon	Argon gas selected

Table 9.3-7 Selection of plasma system process gases status messages

Material Thickness Parameter

Message		Description
0	mil	Thickness measured in 1000th of an inch
1	um	Thickness measured in microns (SI)
2	gsm	Thickness in grams/square metre, used in the paper industry as a standard.
3	gauge	Equivalent to 1 hundredth of a mil

Table 9.3-8 Material thickness parameter status messages

Drive System Status

Message		Description
	OK	Everything OK
1	Emergency stop	Emergency stop situation identified. Generally caused by the pressing of one of the stop buttons on the machine.
2	I/O Fault	Network communications fault, on either the profinet/profibus or Asi systems.

Table 9.3-9 Drive System Status Messages

Message		Description
3	Web break	Drive system has registered a web break situation when there is low tension identified on the substrate.
4	Fast stop	Activated when a drive system component fails or a web break is identified.
5	Watchdog timeout	Communications network monitoring function identifies issues.
6	Drive PLC rack fault	Drive system plc module in fault condition. Contact BML for assistance.
7	Layarms positioning fail	The layarm system has not moved to the correct position. A diagnostic message code maybe shown on the drive diagnostics screen.
8	Tension up fail	Tension function has been activated, but system has not measured correct tension. Check that substrate is passing around the winding mechanism correctly.
9	Drive PLC rack fault	Drive system plc module in fault condition. Contact BOBST for assistance.
10	Shutter drive fault	N/A
11	Spreader drive fault	N/A
12	Strobe drive fault	Viewing strobe drive fault recognised by the machine control system. Investigate the unit located in electrical panel E0_3_2
20	Unwind drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
21	Unwind Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
22	Unwind blower fault	Motor forced cooling fan MCB tripped. Investigate and reset.
23	Unwind over diameter	The drive system has calculated that the diameter is outside the configured parameters.
30	Rewind drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
31	Rewind Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
32	Rewind blower fault	Motor forced cooling fan MCB tripped. Investigate and reset.
33	Rewind over diameter	The drive system has calculated that the diameter is outside the configured parameters.

Table 9.3-9 Drive System Status Messages

Message		Description
40	DTR1 drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
41	DTR1 Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
42	DTR1 blower fault	Motor forced cooling fan MCB tripped. Investigate and reset.
43	DTR1 slipping	When operating the drive in “torque mode” and the roller is reaching its over speed set point, this message will be shown. Reduce the torque set point.
50	DTR2 drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
51	DTR2 Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
52	DTR2 blower fault	Motor forced cooling fan MCB tripped. Investigate and reset.
53	DTR2 slipping	When operating the drive in “torque mode” and the roller is reaching its over speed set point, this message will be shown. Reduce the torque set point. Check that the surface of the high friction cover has not degraded.
60	Drum drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
61	Drum Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
62	Drum blower fault	Motor forced cooling fan MCB tripped. Investigate and reset.
63	Drum slipping	N/A
70	Unwind layarm drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
71	Unwind layarm Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
72	Unwind layarm move to reel fault	Indicated when unable to reach the roll/core position.
73	Unwind layarm move to home fault	Indicated when unable to reach the home position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen

Table 9.3-9 Drive System Status Messages

Message		Description
74	Unwind layarm diameter measure fail	Indicated when unable to reach the roll diameter position
80	Rewind layarm drive fault	Drive fault identified; refer to the drives diagnostic screen for error code and vendor manual for possible reasons.
81	Rewind layarm Profibus fault	Drive module is not recognised on the communications network. Check cabling to the drive unit.
82	Rewind layarm move to reel fault	Indicated when unable to reach the roll/core position.
83	Rewind layarm move to home fault	Indicated when unable to reach the home position. Possible Reasons:- • Mechanical restriction – visually check for obstructions. • Home switch not operating correctly. Monitor indicator led on switch, if induction type • Activation striker not activating the switch – visually check. • Wiring issue – Check signal at plc input • Drive failure – Check diagnostic screen
84	Rewind layarm diameter measure fail	Indicated when unable to reach the roll diameter position
90	Drum spreader drive fault	N/A
91	Drum spreader Profibus fault	N/A
92	Drum spreader blower fault	N/A
93	Drum spreader slipping	N/A

Table 9.3-9 Drive System Status Messages

9.4 Winding Cart Linear Traverse Drive System

The winding cart is moved into the vacuum chamber to allow the evaporation process to take place. It is moved out of the chamber to allow access to the evaporation source area and winding area for cleaning and maintenance.

A chain driven shaft and motor gearbox is located at the winding cart. This moves the winding cart in to or out of the vacuum chamber.

The motor has an inbuilt brake that prevents any movement of the cart moving if the power is not applied.

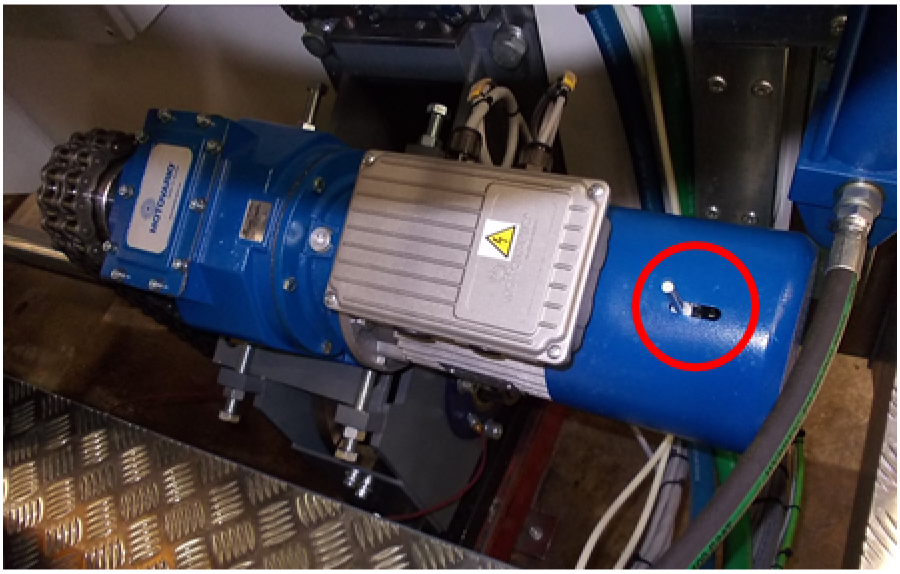

Figure 9.4-1 Motor and gearbox with the inbuilt brake

Proximity switches are located inside/on the chamber. These activate when the winding cart is at the open or closed position and stop it moving any further.

Figure 9.4-2 Winding cart location indicating limit sensors – slow open and fully open

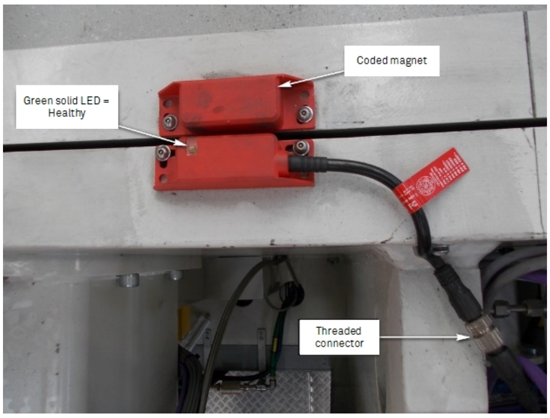

Figure 9.4-3 Winding cart location indicating limit sensor – Fully closed activation – located on top of the chamber.

Two additional switches change the speed that the winding cart moves when it approaches the open or closed position. Activation plates are located on the upper tracks to activate the switches.

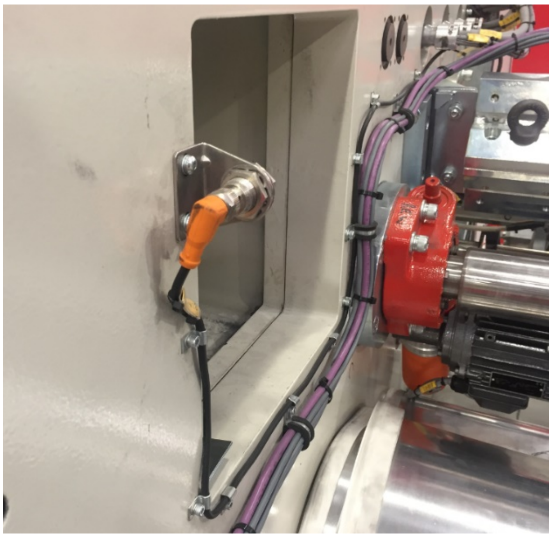

Figure 9.4-4 Winding cart location indicating limit sensor – slow close activation

The open, closed and enable switches on the HMI are used to move the winding cart in to or out of the vacuum chamber.

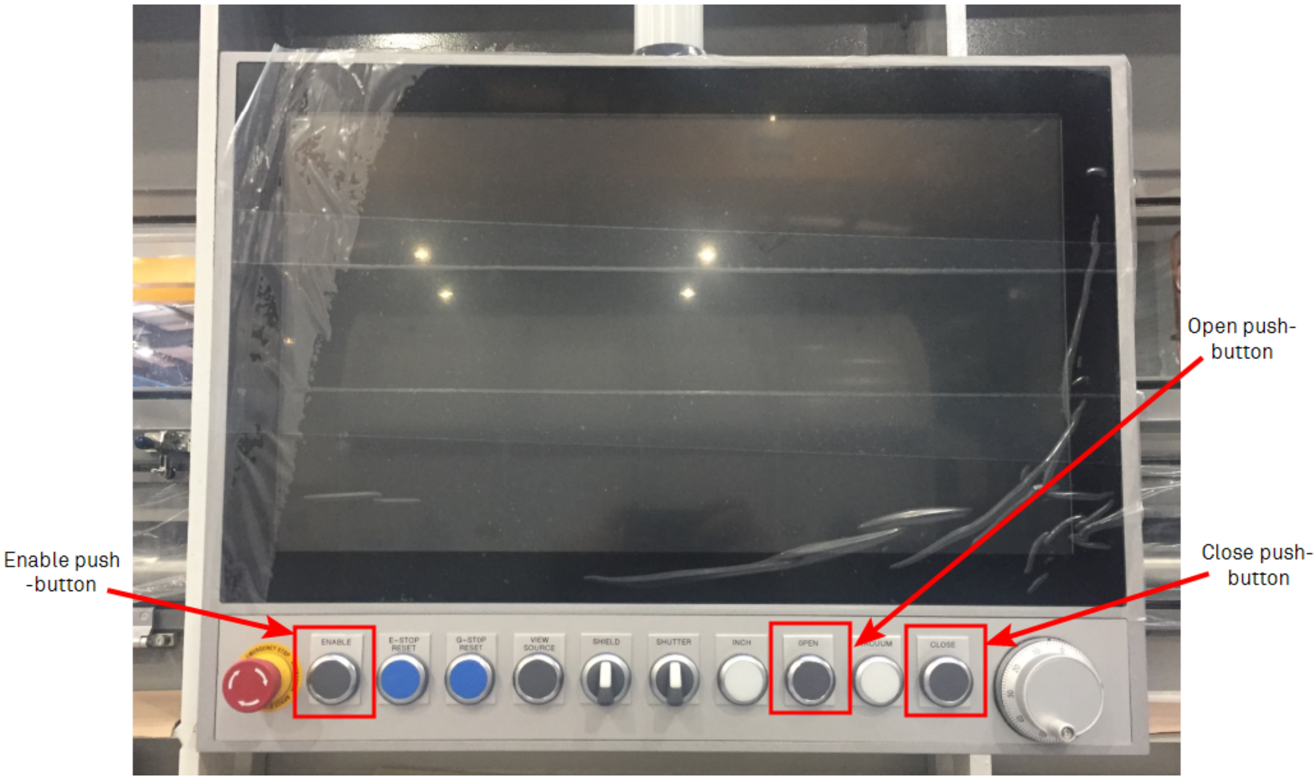

Figure 9.4-5 Winding cart traverse controls on the operator interface panel

	Warning !
	Make sure that there are no obstructions behind the winding cart before moving it out of the vacuum chamber. Make sure that the vacuum chamber is empty before moving the winding cart in.

The open and close buttons on the HMI cannot be activated until the enable button is pressed.The open and close buttons on the HMI cannot be activated until the mech enable button is pressed. This button is located on the rear HMI.

Figure 9.4-6 The traverse enable push-button at the rear of machine

There is a 5-second delay after the open or closed button is pressed and before the cart starts to move. During this time, the warning lights start to flash and siren operates. These indicate that the cart is about to move and provide a warning to any people in the area. These warning indicators will remain on until the cart has reached the fully open or closed position.

Moving the winding cart out of the vacuum chamber

The winding cart will not move in to the chamber when the traverse movement lockout isolator is on.

	Warning !
	Check that there are no obstructions behind the winding cart before moving it out of the chamber.

To move the winding cart out of the chamber;-

1. Make sure that there are no obstructions behind the winding cart
2. Turn the traverse movement lockout isolator on the E06 panel to the off position
3. Press the enable button on the panel at the rear of the machine.Press the mech enable button on the rear HMI.
4. Make sure that there are no obstructions behind the winding cart
5. Press the enable and open button on the HMI
    - The warning lights will flash and the siren will sound
    - The winding cart will start to move after a 5 second delay
    - The winding cart will slow down when it is close to the fully open position
    - The winding cart will stop at the fully open position

Moving the winding cart in to the vacuum chamber

The winding cart will not move in to the chamber when the following interlocks are active:

- Shutter Closed
- Coating shield raised
- Winding cart water is OK
- Evaporation enclosure water is OK
- Enable push-button is first initially pressed.Or, the mech enable push-button is first initially pressed.
- Electrical protection to the drive unit is healthy

If any interlock is active the warning lights will not flash and the siren will not sound when the close button is pressed. The interlocks can be checked on the diagnostic screen on the HMI by selecting the traverse in or out interlock logic.

	Warning !
	Only one operator should be responsible for closing the machine. This operator must check that the vacuum chamber is empty before pressing the enable button to move the winding cart. **Keep hands clear of the opening between the winding cart and vacuum chamber when the winding cart is moving.**

To move the winding cart in to the chamber;-

1. Make sure that the vacuum chamber is empty
2. Press the enable button on the panel at the rear of the machine.Or, press the mech enable button on the rear HMI.
3. Make sure that the vacuum chamber is empty
4. Press the enable and close button on the HMI
    - The warning lights will flash and the siren will sound
    - The winding cart will start to move after a 5 second delay
    - The winding cart wil slow down when it is close to the fully closed position
    - The winding cart will stop at the fully open position
5. Turn the traverse movement lockout isolator on the E06 panel to the on position

Locking Out the Traverse Motor

Warning !

The movement of the winding cart should be isolated when working inside the vacuum chamber by locking out the movement lockout isolator

To lockout the traverse movement lockout isolator, turn it through 90 degrees and place a padlock on the switch. This action also disables the winding mechanism drive system

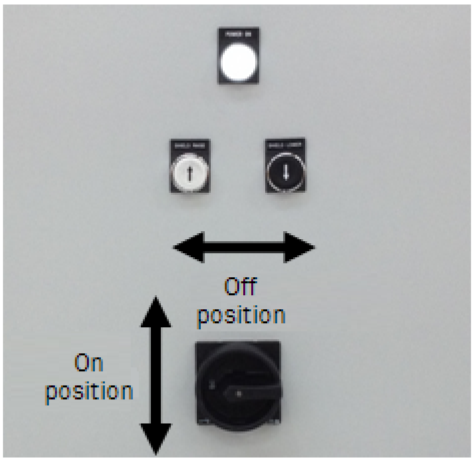

Figure 9.4-7 The traverse movement isolator.

9.5 Chamber Windows

Clear profiled polycarbonate material is provided on all viewing windows on the chamber to allow viewing. This material is a special optical grade and Bobst Manchester Ltd can supply replacements.

Figure 9.5-1 Example of an operator checking windows.

The evaporation source viewing window has pieces of toughened glass placed in front of it in the Vacuum Chamber to allow cleaning.

	**Warning !**
	Do not remove or modify the supplied viewing windows

See chapter Machine Cleaning on page 1 of this manual in regard to cleaning of the windows.

	**Warning !**
	Do not use any type of solvents/acids to clean the window. **As there will be chemical erosion, that could lead to failure of the window.**

There are spacers provided to allow the location of the windows onto the chamber and to spread the loading between the fastener and the window and are used to minimise damage to the window, by “overtightening” the fasteners onto the polycarbonate sheet.

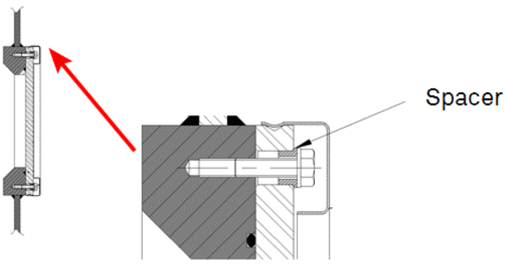

Figure 9.5-2 Cross Section View of Chamber Rear Window.

Caution !

Ensure Spacers are fitted between each location bolt.

The bolts should be tightened using fingers only. When a vacuum is pulled on the chamber, they should be then be finger tightened again, as the window will be pulled onto the mating surface.

Do not “overtighten” the holding/location bolts on the windows.

Undue pressure will cause the Polycarbonate to fail and shatter or crack.

If unsure about replacing the windows, seek advice from Bobst Manchester Ltd.

Rectangular Windows

The bolts should be tightened in the following sequence (this is just an example);-

T1, B16, T16, B1, T8, B9, T2, B15, T15, B2, L1, L2, T9, B8, T4, B12, B5, T11, T5, T3, B14, T11, B3, T14, B10, T6, T13, B11, B7, B4, B13, B6, T12, T7, T10,

Figure 9.5-3 Tightening Bolts Pattern - Rectangular Window

Rear Square Windows

The bolts should be tightened in the following sequence; - 1, 5,3,7,2,6,4,8

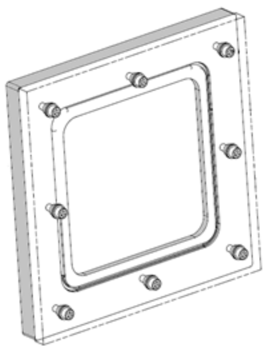

Figure 9.5-4 Tightening bolts pattern - square window.

If unsure about replacing the windows, seek advice from Bobst Manchester Ltd.

9.6 HMI Touch Screen

The human machine interface (HMI) is the main panel that the operator will use to control the machine. This panel has a touch sensitive screen that allows touch selection.

Boot Screen

After switching on the main isolator, the computer will power up and display the machine boot screen.

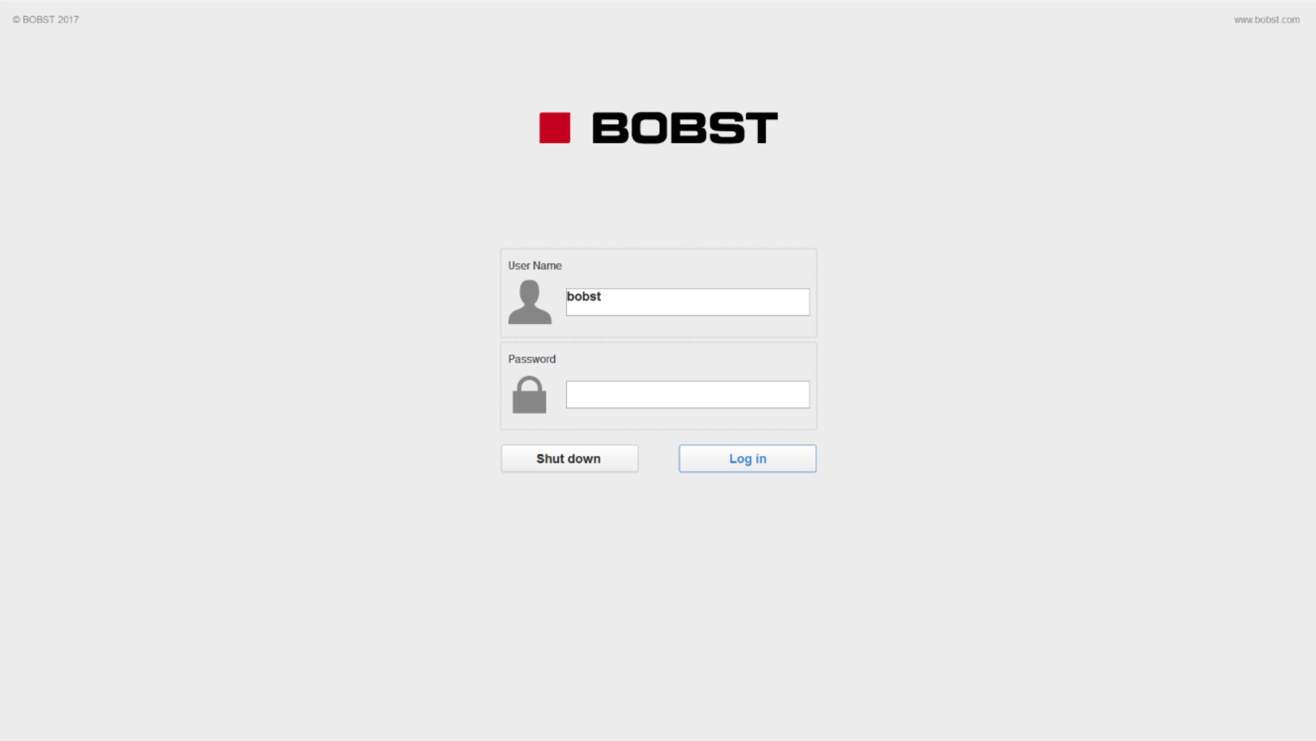

Figure 9.6-1 Machine boot screen.

Normally the program should be operated in an operator mode.

Touch the screen or use the keyboard and mouse to proceed.

All other normally used screens are accessible from this screen.

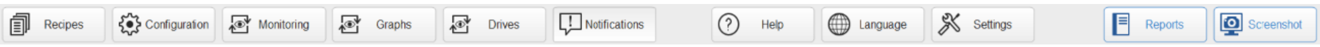

Figure 9.6-2 HMI Menu Selection Sections.

To change the screen, touch the relevant section on the screen.

Note !

The access to the higher-level password will be given to the customers designated manager at the time of machine start up.

This cannot be changed, as the HMI software does not allow it.

Soft Keyboard

An on screen keyboard allows data entry.